Structured Gold Forward with Knock-Out Barriers

Product Development Memorandum

Derivatives Structuring Desk

February 2026

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CONFIDENTIAL

TO: Product Committee, Alphabank S.A.

FROM: Derivatives Structuring Desk

DATE: February 2026

RE: Pricing and Risk Analysis — Zeus Gold Group AG Hedging Facility

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MARKET UPDATE — February 26, 2026

This memorandum has been updated with market data as of February 26, 2026. Gold has moved to $5,203/oz (from $4,900 on Feb 1), EUR/USD to 1.181, and realized gold volatility has risen to 41%. The position reversal is now firmly established with Z Group PV at EUR +64M.

Executive Summary

This memorandum presents our analysis of a proposed structured hedging facility for Zeus Gold Group AG ("Z Group"). The product combines exposure to LBMA gold prices with automatic termination features linked to EUR/USD exchange rate movements.

We have developed a comprehensive pricing framework validated through multiple methodologies. Our analysis identifies several structural considerations that warrant discussion before proceeding to term sheet finalization.

Transaction Summary

Parameter	Specification
Notional Principal	EUR 500,000,000
Reference Asset	LBMA Gold PM Fixing (USD/oz)
Strike Price	USD 4,600 per troy ounce
Tenor	2 years (March 2026 — February 2028)
Knock-Out Barriers	EUR/USD < 1.05 or EUR/USD > 1.25
Settlement	Cash settled in EUR at prevailing EUR/USD rate

Key Findings

Metric	Result
Z Group Present Value	EUR +64 million
Alphabank Present Value	EUR −64 million

Knock-Out Probability	64%
Expected Contract Duration	12 months
Forward Moneyness	113.1%

The positive present value for Z Group reflects gold's surge above the strike price, placing the forward at 113% moneyness. The 64% knock-out probability is driven primarily by the upper barrier's proximity to current EUR/USD spot (5.8% distance), representing a fundamental reversal from earlier market conditions where the lower barrier dominated.

1 Transaction Overview

Zeus Gold Group, a Frankfurt-headquartered jewelry manufacturer, seeks to hedge its USD-denominated gold procurement costs while managing EUR/USD translation risk. The proposed facility would run for two years commencing March 2026.

1.1 Settlement Mechanics

At settlement time τ (maturity or knock-out, whichever occurs first), with LBMA gold fixing at price P:

$$Z\ Group\ Payoff = N \times (P\tau - K) / K$$

$$Alphabank\ Payoff = N \times (K - P\tau) / K$$

where:

- N = 500,000,000 EUR (notional principal)
- K = 4,600 USD/oz (strike price)
- Pτ = LBMA Gold PM fixing at settlement

1.2 Knock-Out Mechanism

The contract terminates immediately upon the first occurrence of:

$$X_t < L \ \ or \ \ X_t > U$$

where X_t denotes the EUR/USD rate, L = 1.05 (lower barrier), and U = 1.25 (upper barrier).

The stopping time is defined as:

$$\tau_{ko} = inf\{t \geq 0 : X_t \notin (L, U)\}$$

Settlement occurs at τ = min(T, τ_{ko}) where T = 2 years.

2 Mathematical Framework

2.1 Stochastic Model

Both underlying assets follow geometric Brownian motion under the risk-neutral measure $\mathbb{Q}$.

Gold Price Dynamics (USD):

$$dS_t/S_t = (r_USD - q)\, dt + \sigma_S\, dW_t^{(1)}$$

EUR/USD Exchange Rate:

$$dX_t/X_t = (r_EUR - r_USD)\, dt + \sigma_X\, dW_t^{(2)}$$

Correlation Structure:

$$^{O}[dW_t^{(1)} \cdot dW_t^{(2)}] = \varrho\, dt$$

2.2 Parameter Estimates

Parameter	Symbol	Value	Source
Gold spot	S_0	USD 5,203/oz	yfinance live (GC=F, Feb 2026)
EUR/USD spot	X_0	1.181	yfinance live (EURUSD=X, Feb 2026)
USD risk-free rate	r_USD	3.6%	13-week T-bill (^IRX)
EUR risk-free rate	r_EUR	2.0%	ECB deposit rate (configured)
Gold volatility	σ_S	41%	EWMA (λ=0.94, GC=F)
EUR/USD volatility	σ_X	6.2%	EWMA (λ=0.94, EURUSD=X)
Correlation	ρ	−0.30	126-day rolling (fallback)
Gold convenience yield	q	3.6%	Futures term structure (GC=F vs GCJ26.CMX)

2.3 Risk-Neutral Valuation

The present value under the EUR risk-neutral measure:

$$V_0 = {}^{O\varphi}[e\text{^}(-r_EUR \cdot \tau) \cdot Payoff\tau]$$

The path-dependent barrier feature precludes closed-form solutions, necessitating Monte Carlo methods.

3 Numerical Implementation

3.1 Simulation Methodology

We employ Monte Carlo simulation with the following specifications:

Parameter	Value	Rationale
Simulation paths	100,000	Adequate precision for indicative pricing
Time steps	504	Daily monitoring over 2 years
Random seed	Fixed	Reproducibility
Variance reduction	Antithetic + Control	~60% error reduction

Discretization Scheme

Asset prices are simulated using the exact log-normal solution:

$$S_{t+\Delta t} = S_t \cdot exp[(\mu_S - \sigma_S^2/2)\Delta t + \sigma_S\sqrt{\Delta t} \cdot Z_1]$$

$$X_{t+\Delta t} = X_t \cdot exp[(\mu_X - \sigma_X^2/2)\Delta t + \sigma_X\sqrt{\Delta t} \cdot Z_2]$$

where Z_1, Z_2 are correlated standard normals generated via Cholesky decomposition:

$$Z_2 = \varrho\, Z_1 + \sqrt{(1-\varrho^2)}\, Z\perp$$

3.2 Variance Reduction

Two techniques are implemented to improve computational efficiency:

Antithetic Variates: For each path with innovations $\{Z_t\}$, we also simulate the reflected path $\{-Z_t\}$. The negative correlation between paired paths reduces variance.

Control Variate: The vanilla gold forward (without barriers) serves as a control:

$$\hat{V}_adj = \hat{V}_exotic + \beta(V_vanilla\text{^}analytical - \hat{V}_vanilla)$$

where β is the optimal control coefficient estimated from sample covariance.

Combined, these techniques reduce standard errors by approximately 60%.

4 Pricing Results

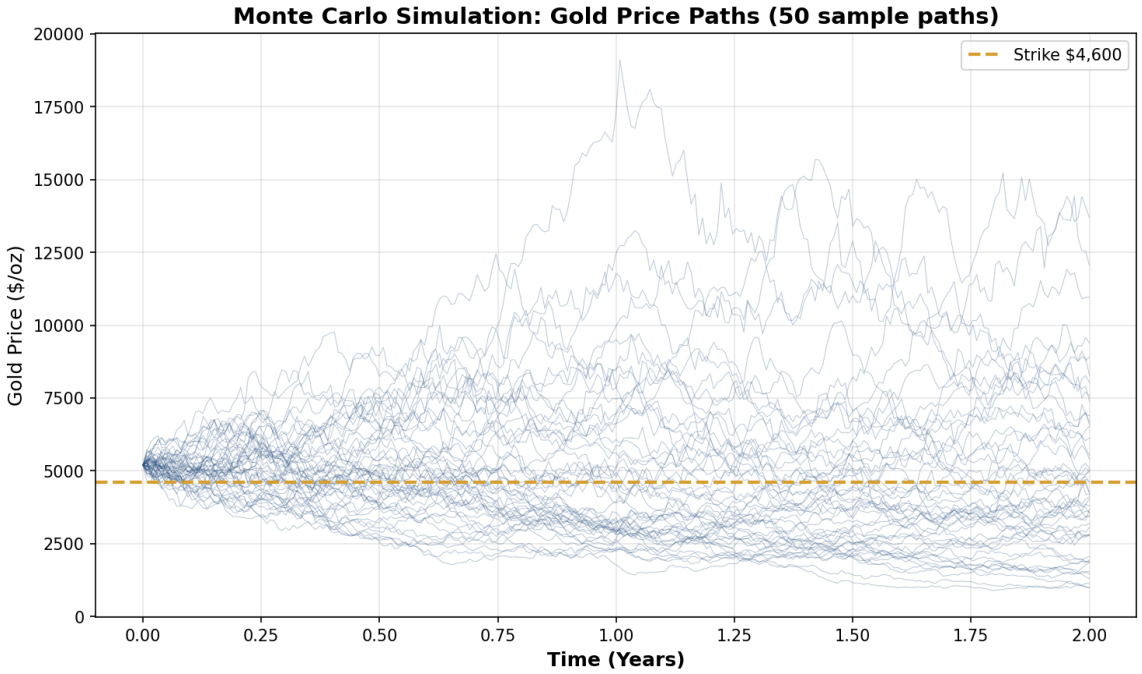
4.1 Base Case Valuation

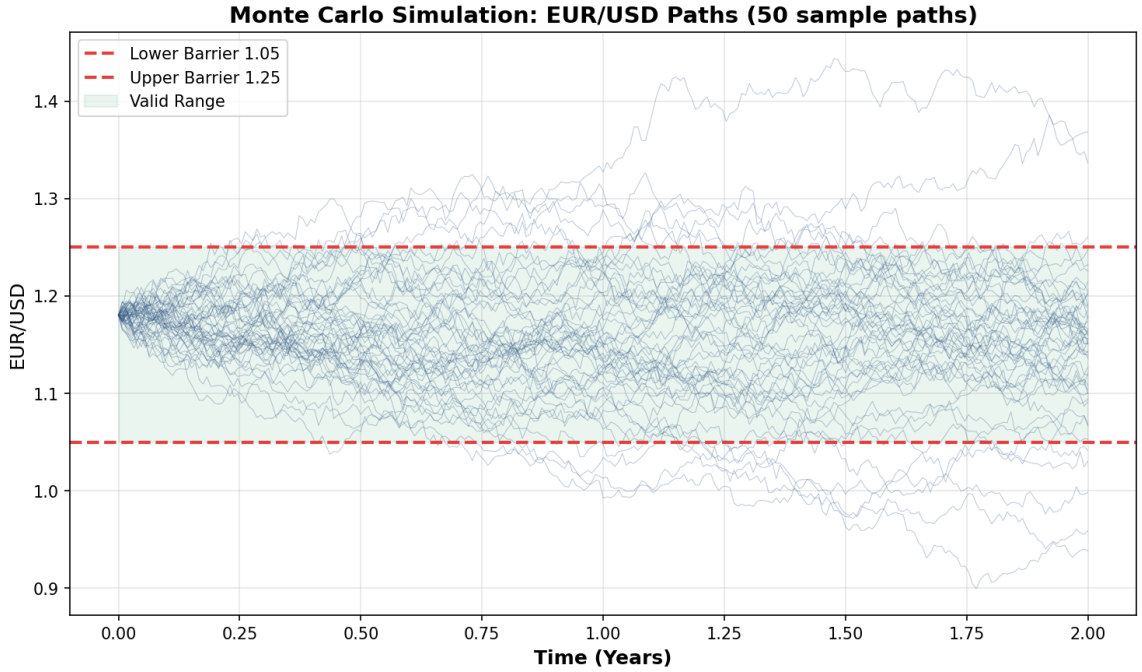
Metric	Value
Z Group Present Value	EUR +63,968,230
Alphabank Present Value	EUR −63,968,230
Standard Error	EUR 896,173
95% Confidence Interval	[+62.2M, +65.7M]

4.2 Barrier Analysis

Metric	Value
Overall Knock-Out Rate	64.2%
Lower Barrier Breaches	27.3%
Upper Barrier Breaches	36.9%
Average Time to Knock-Out	0.98 years (11.8 months)

The barrier breach profile has shifted dramatically compared to earlier market conditions. The upper barrier now accounts for the majority of knock-outs (37% vs 27% lower), reflecting the EUR/USD spot at 1.181—only 5.8% from the upper barrier versus 11.1% from the lower. The interest rate differential ($r_{EUR} - r_{USD} = -1.6\%$) still implies euro depreciation drift, but the proximity asymmetry dominates.





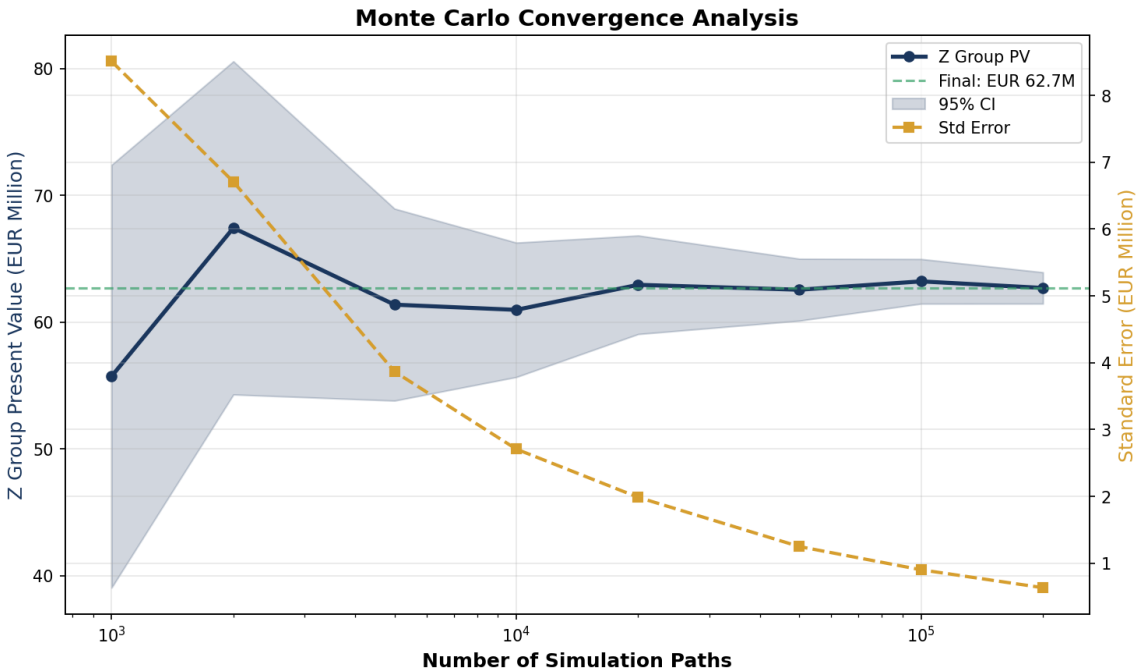
4.3 Convergence Verification

Monte Carlo estimates stabilize as path counts increase:

Paths	Price Estimate	Standard Error
5,000	EUR +61.5M	EUR 3,844K
10,000	EUR +61.6M	EUR 2,698K

25,000	EUR +62.7M	EUR 1,768K
50,000	EUR +63.1M	EUR 1,241K
100,000	EUR +64.0M	EUR 896K

Standard errors decay proportionally to 1/√n, confirming proper convergence behavior.



5 Critical Assessment

5.1 Strike Price Analysis

The specified strike of USD 4,600/oz warrants careful examination.

Forward Price Calculation:

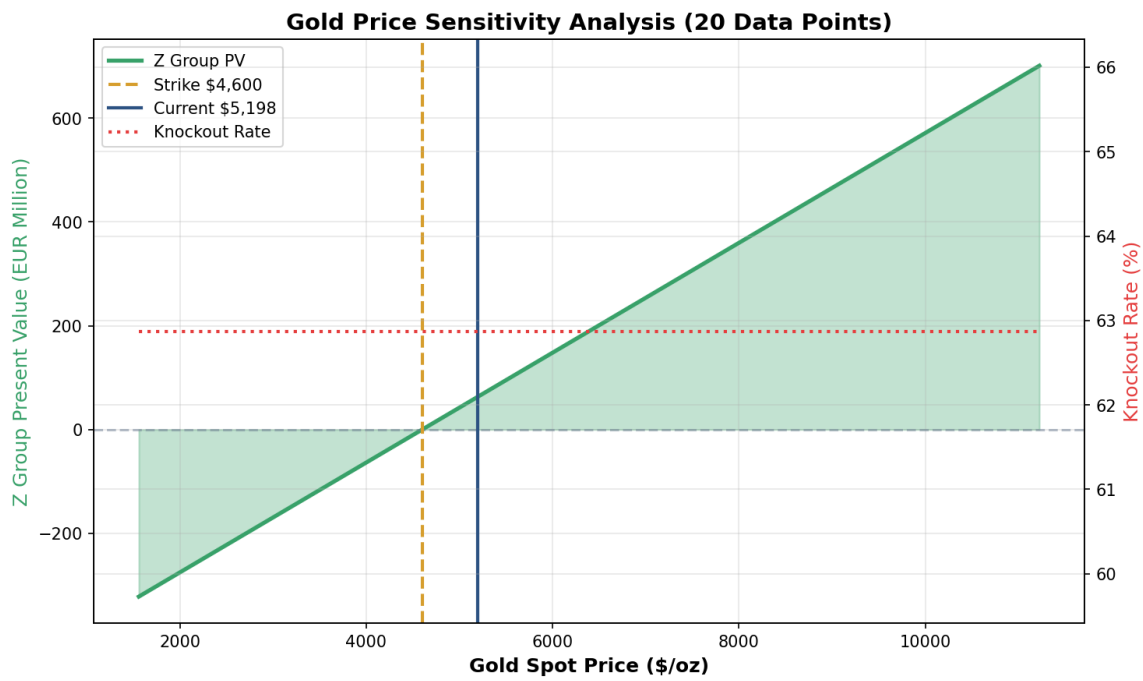
$$F_0,T = S_0 \cdot e^{((r_USD - q) \cdot T)} = 5{,}203 \cdot e^{((0.036-0.036)\cdot 2)} \approx USD\ 5{,}203/oz$$

With convenience yield approximately equal to the USD risk-free rate, the forward is near spot. The strike sits 12% below the forward, placing Z Group in the money:

$$Moneyness = F_0,T / K = 5{,}203 / 4{,}600 = 113.1\%$$

Alternative Strike Analysis:

Strike	Forward Relationship	Z Group PV
USD 4,000	23% below forward	EUR +146M
USD 4,300	17% below forward	EUR +101M
USD 4,600	12% below forward	EUR +63M
USD 4,900	6% below forward	EUR +29M
USD 5,200	At-the-money	EUR ~0M
USD 5,500	6% above forward	EUR −27M



5.2 Barrier Configuration

With EUR/USD at 1.181, the upper barrier at 1.25 is now the proximate risk:

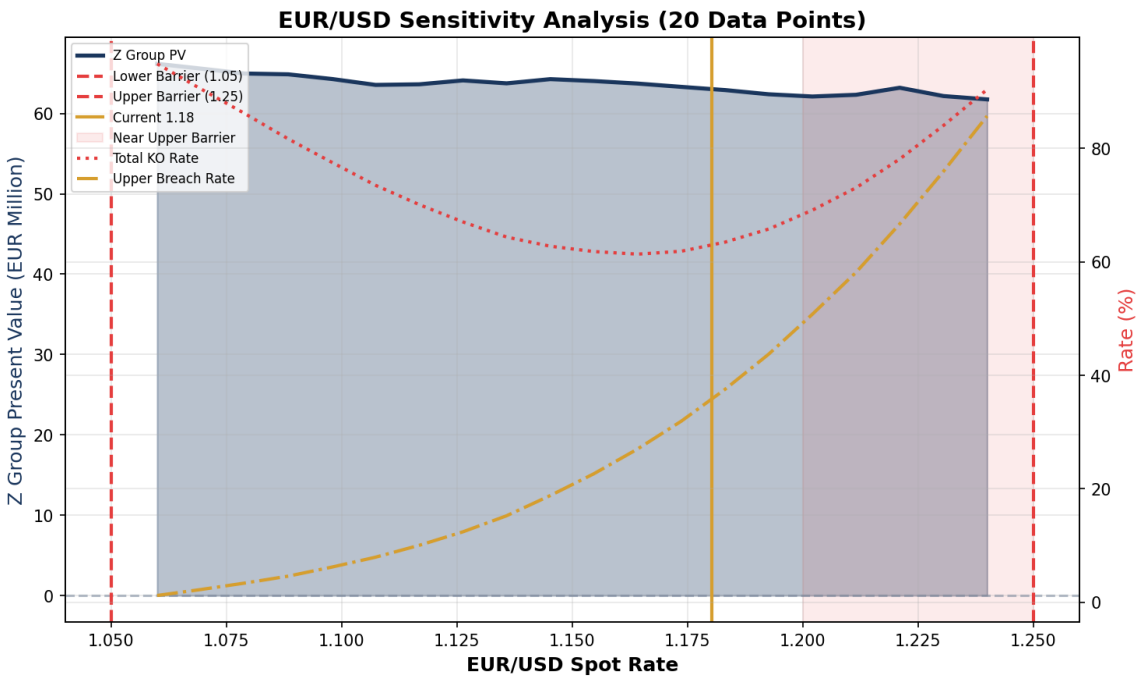
$$Distance\ to\ Upper\ Barrier = (U - X_0) / X_0 = (1.25 - 1.181) / 1.181 = 5.8\%$$

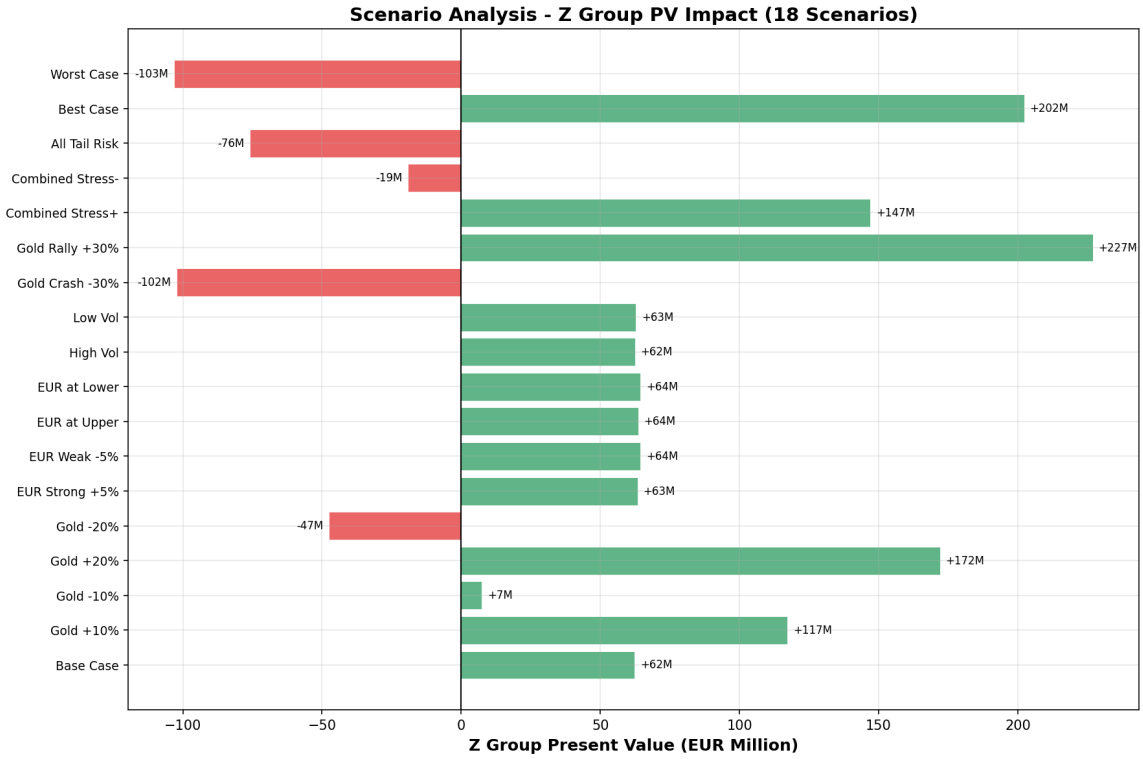
$$Distance\ to\ Lower\ Barrier = (X_0 - L) / X_0 = (1.181 - 1.05) / 1.181 = 11.1\%$$

With 6.2% annual EUR/USD volatility, the asymmetric positioning creates an upper-barrier-dominated knock-out profile.

Alternative Configurations:

Corridor	Knock-Out Rate	Expected Duration
[1.05, 1.25]	64%	12 months
[1.00, 1.30]	27%	15 months
[0.95, 1.35]	10%	17 months

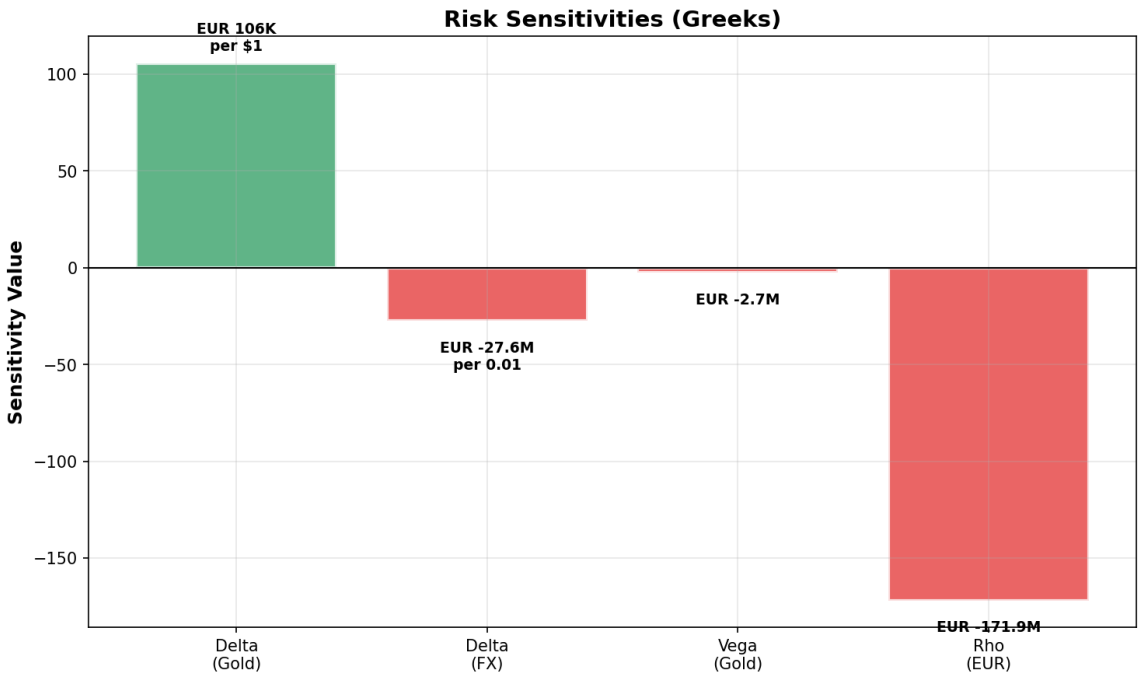



Scenario Analysis - Z Group PV Impact (18 Scenarios)
Worst Case
Best Case
All Tail Risk
Combined Stress-
Combined Stress+
Gold Rally +30%
Gold Crash -30%
Low Vol
High Vol
EUR at Lower
EUR at Upper
EUR Weak -5%
EUR Strong +5%
Gold -20%
Gold +20%
Gold -10%
Gold +10%
Base Case
-103M
+202M
-76M
-19M
+147M
+227M
-102M
+63M
+62M
+64M
+64M
+64M
+63M
-47M
+172M
+7M
+117M
+62M
−100
−50
0
50
100
150
200
Z Group Present Value (EUR Million)

6 Risk Sensitivities

6.1 Greeks Summary

Greek	Value	Interpretation
Delta_gold	EUR 105,677 per USD 1	First-order gold sensitivity
Gamma_gold	EUR −669	Gold convexity
Delta_FX	EUR −8.18M per 0.01 FX	EUR/USD sensitivity (negative: upper barrier risk)
Vega_gold	EUR −2.88M per 1% vol	Gold vega (higher vol increases KO probability)
Rho_EUR	EUR −1,031M per 1bp	EUR rate sensitivity
Rho_corr	EUR −973K per 0.05 corr	Correlation sensitivity



6.2 Hedging Implications

Delta Hedging: The gold delta of EUR 106K per dollar implies a hedge ratio of approximately:

$$Gold\ Hedge = (\Delta_gold / S_0) \times K = (105{,}677 / 5{,}203) \times 4{,}600 \approx 93{,}400\ oz$$

Barrier Risk: With EUR/USD at 1.181, the upper barrier at 1.25 is only 5.8% away. As EUR/USD approaches either barrier, gamma and delta become increasingly unstable—the characteristic “pin risk” of barrier options. The negative FX delta (EUR −8.2M per 0.01) reflects that EUR appreciation toward 1.25 destroys contract value through knock-out.

7 Model Validation

7.1 Alternative Specifications

To ensure robustness, we compared valuations across three model specifications:

Model	Z Group PV	Knock-Out Rate
Base GBM	EUR +63.1M	64.5%
Heston Stochastic Vol	EUR +64.1M	64.1%
Merton Jump-Diffusion	EUR +65.7M	64.1%

All models converge within 4%, with Heston and Merton producing slightly higher valuations due to stochastic volatility and jump dynamics amplifying the in-the-money payoff. Model specification risk remains secondary to market parameter uncertainty.

7.2 Analytical Benchmark

The vanilla gold forward (without barriers) provides a sanity check:

$$V_vanilla = e^{(-r_EUR \cdot T)} \cdot N \cdot (F_0, T - K) / K = e^{(-0.020 \times 2)} \cdot 500M \times (5{,}203 - 4{,}600) / 4{,}600$$

$$V_vanilla = EUR\ +63{,}004{,}791$$

The knock-out version (EUR +64.0M) is EUR 1.0M higher than the vanilla, reflecting that early termination locks in profits when gold is above strike—a reversal from the prior regime where knock-outs destroyed value.

8 Conclusions and Recommendations

8.1 Summary of Findings

The proposed structure is technically sound and priceable using standard Monte Carlo techniques. The February 2026 market environment has fundamentally altered the risk profile:

1. Position reversal: Gold's surge to $5,203/oz places Z Group firmly in the money (113% moneyness), with a positive PV of EUR +64M. This contrasts sharply with earlier market conditions where the position was deeply out-of-the-money.

2. Barrier risk shift: The upper EUR/USD barrier at 1.25 is now only 5.8% from spot, making it the primary knock-out driver (37% of paths). The 64% total knock-out rate and 12-month expected duration represent a more balanced risk profile than the prior 93%/5-month scenario.

3. Elevated volatility: Gold volatility at 41% (EWMA) is more than double historical norms, increasing both the potential upside for Z Group and the mark-to-market volatility for Alphabank's hedging book.

8.2 Recommendations

We recommend proceeding to term sheet stage with the following considerations:

1. Alphabank should carefully assess credit exposure given the positive Z Group PV, ensuring adequate collateral arrangements

2. The upper barrier proximity (5.8%) warrants active monitoring—EUR appreciation events could trigger knock-out and crystallize Z Group's gain

3. Consider whether alternative barrier configurations (e.g., [1.00, 1.30] with 27% KO rate) would provide Z Group with more durable hedging protection

8.3 Next Steps

Upon Committee approval, we will:

- Finalize term sheet documentation
- Establish hedging framework with Trading Desk
- Coordinate credit approval with Risk Management
- Schedule client presentation

[END OF MEMORANDUM]

Appendix A: Nomenclature

Symbol	Description
S_t	Gold spot price (USD/oz) at time t
X_t	EUR/USD exchange rate at time t
K	Strike price (USD 4,600/oz)
N	Notional principal (EUR 500M)
T	Maturity (2 years)
L, U	Lower (1.05) and upper (1.25) barriers
r_EUR, r_USD	Risk-free rates
σ_S, σ_X	Volatilities
ρ	Correlation coefficient
q	Gold convenience yield
τ	Settlement time
τ_KO	Knock-out time

Appendix B: Figure Index

1. Monte Carlo Simulation: Gold Price Paths — gold_paths.png
2. Monte Carlo Simulation: EUR/USD Paths with Barriers — eurusd_paths.png
3. Monte Carlo Convergence Analysis — convergence.png
4. Gold Price Sensitivity Analysis — gold_sensitivity.png
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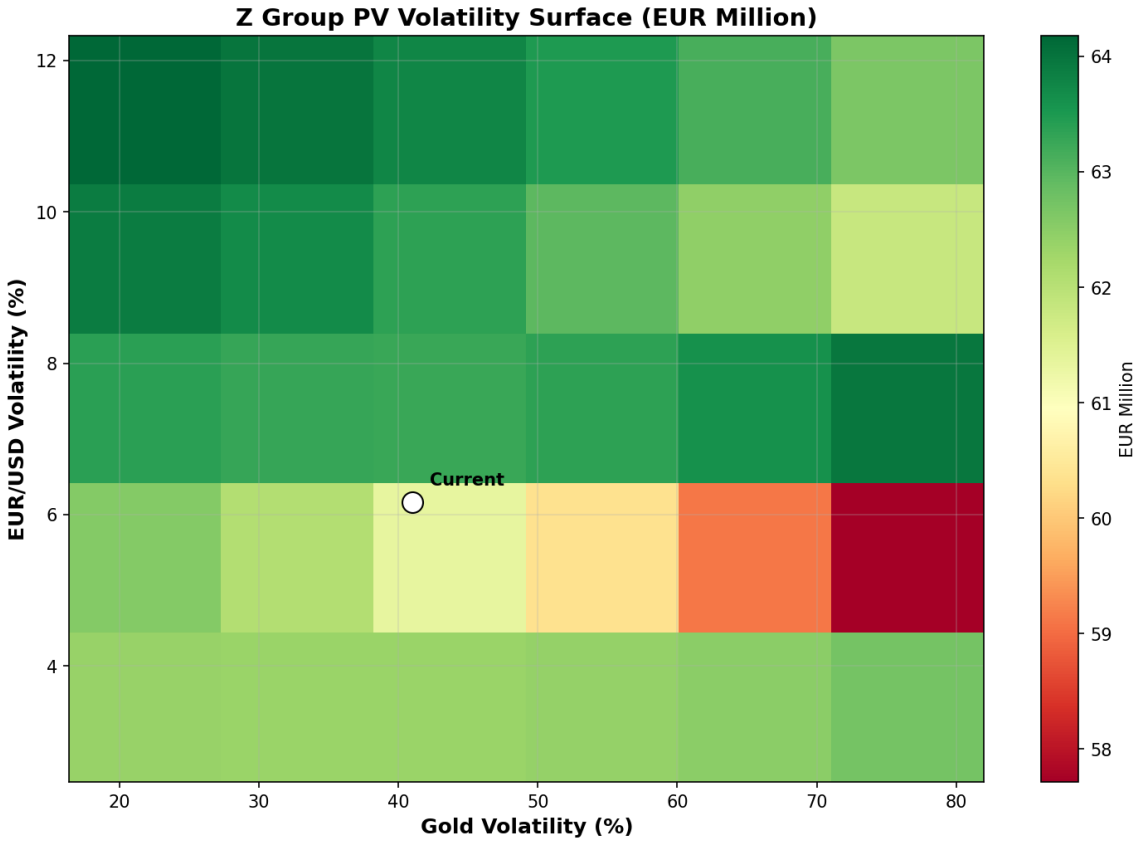
6. Scenario Analysis — Z Group PV Impact — scenario_analysis.png
7. Risk Sensitivities (Greeks) — greeks_chart.png
8. Volatility Surface (2D Grid) — volatility_surface.png
9. Correlation Sensitivity Analysis — correlation_sensitivity.png
10. Payoff Distribution — payoff_distribution.png
11. Knockout Time Distribution and Barrier Breach Analysis — knockout_analysis.png
12. Market Conditions Change (Jan → Feb 2026) — valuation_comparison.png
13. Gold Price Evolution — gold_history.png

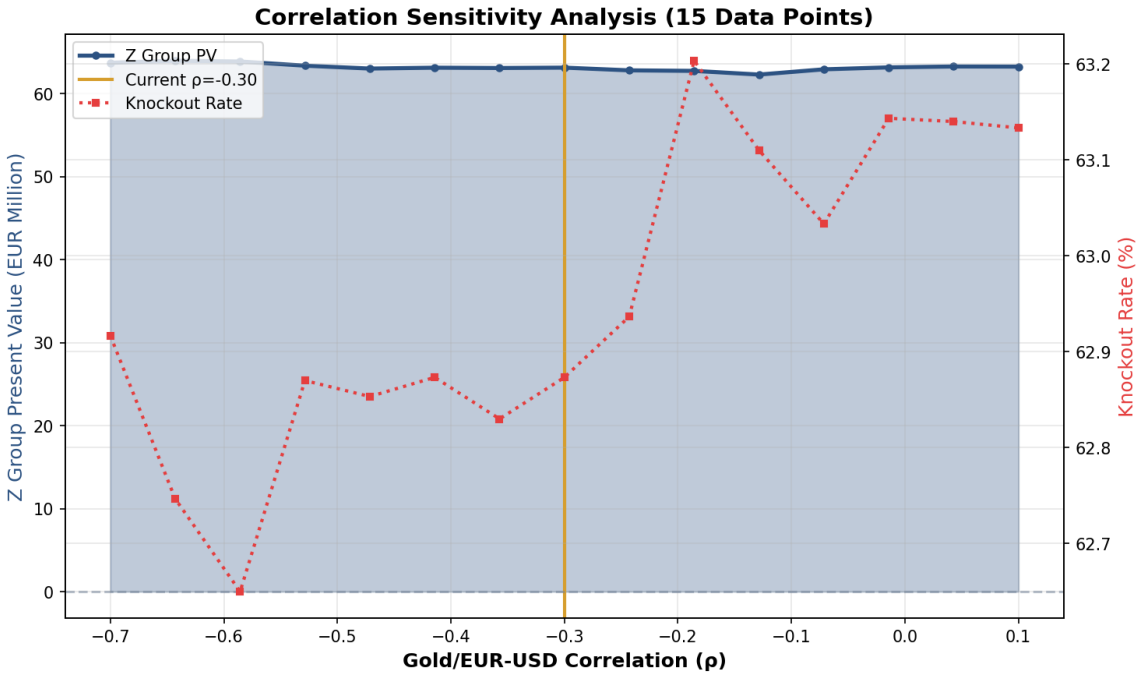
Appendix C: Additional Charts

The following charts provide supplementary analysis supporting the main body of this memorandum.

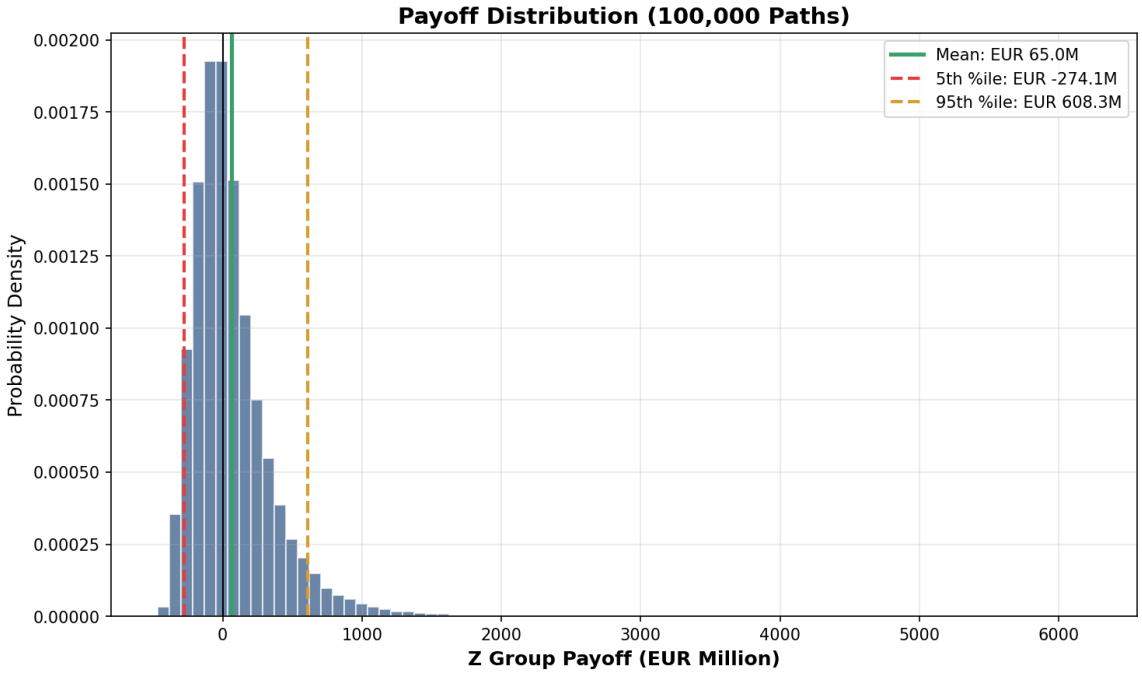
Volatility Surface



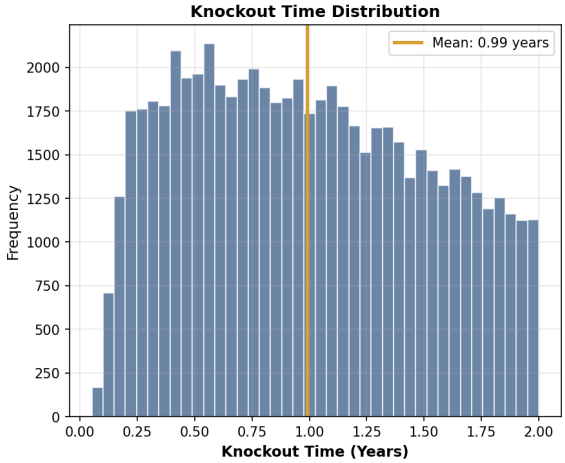
Correlation Sensitivity

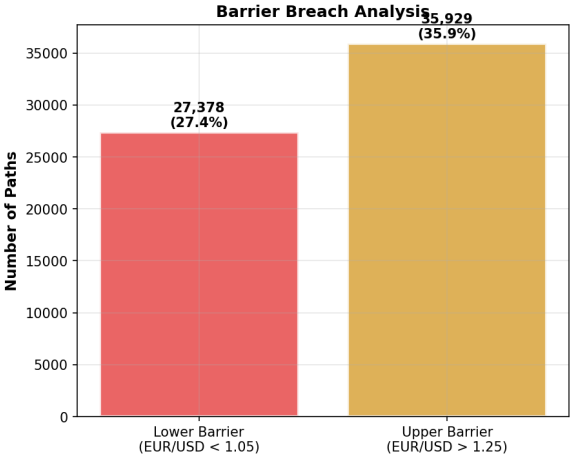


Payoff Distribution

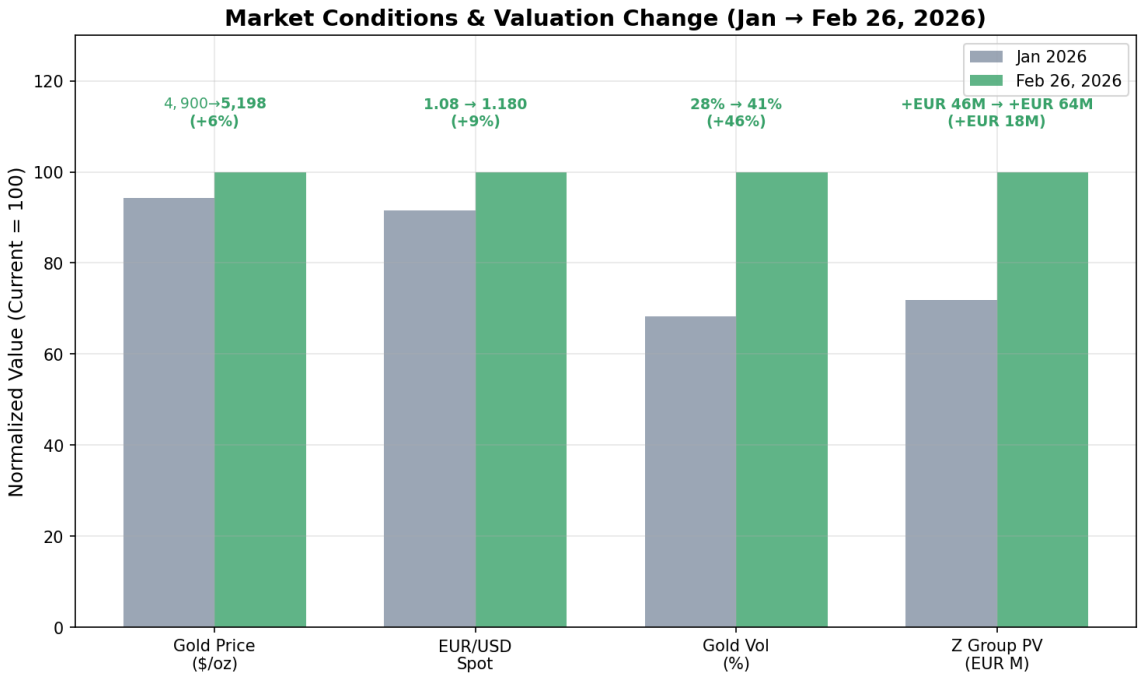


Knockout Analysis

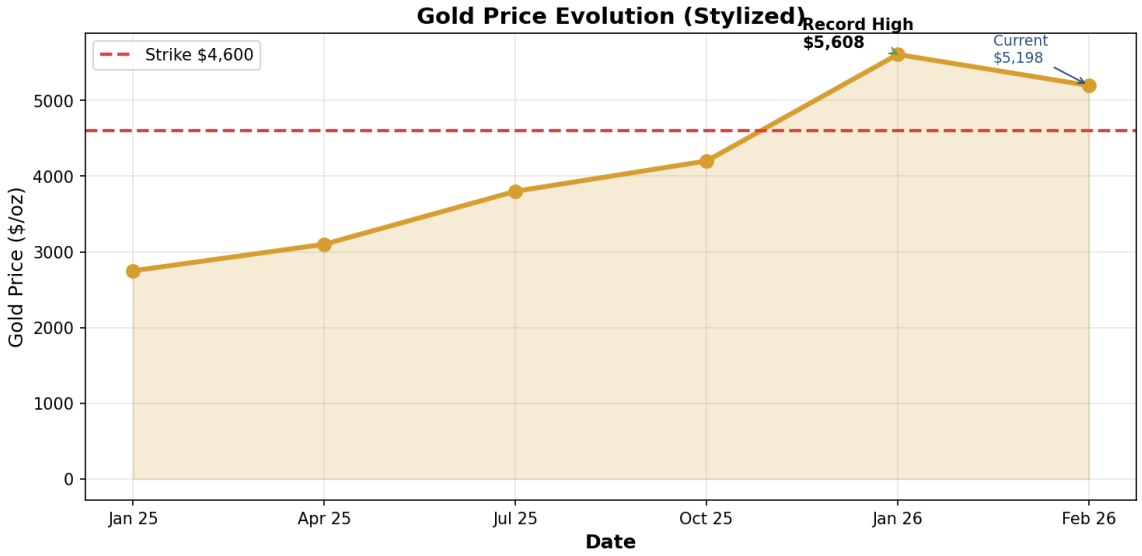




Valuation Comparison: Jan vs Feb 2026



Gold Price History


Gold Price Evolution (Stylized)
Record High
$5,608
Current
$5,198
Strike $4,600
Gold Price ($/oz)
0
1000
2000
3000
4000
5000
Jan 25
Apr 25
Jul 25
Oct 25
Jan 26
Feb 26
Date